

This document provides the background to the proposal of the Hierarchical Self-Retracton Forgiveness (HSRF) model. The model is introduced and discussed in the following paper:

Haghshenas, A. (2025). Self-retraction as redemption: Forgiveness for repentant authors. *Accountability in Research*, 1–14. <https://doi.org/10.1080/08989621.2025.2471513>

Ignored Warnings: A Case Study of Systemic Resistance in Addressing Scientific Misconduct at Shiraz University

During my doctoral studies in 2017, I discovered that some scientific papers and dissertations from our department contained fabricated data. Some experiments were either never conducted, or the reported measurements were not performed as claimed. After consulting with experienced experts and attending the relevant defense meetings, I presented my concerns and initial evidence of misconduct to the supervisory team (two professors) in a confidential and professional manner. I explicitly requested that they refrain from publishing these results. However, my concerns were dismissed, and I was assured that no fraudulent or suspicious findings would be published.

Three years later, in 2020, after I had graduated, I was shocked to find new papers published that reported the same dubious results I had previously warned about. Consequently, I informed the publishers, journals, and the Committee of Scientific Ethics at Shiraz University. Despite my efforts, none of the six papers have been retracted or revised.

1st Case

The Lost Field of Shiraz University

In this case, the results of a two-year field experiment were reported in a paper (Hedayati-Firoozabadi et al., 2020) with the experiment location claimed to be Shiraz University. Having studied in the same department during the claimed experimental seasons, I can confirm that no such experiment was conducted at the School of Agriculture, Shiraz University. Before the dissertation defense, the authors had published another paper in Farsi (Hedayati-Firoozabadi et al., 2017) with considerable overlap in results, but the location was reported as Yazd, Iran; a different province with a completely different climate.

During my 2017 communications with the supervisors, I used satellite images from Google Earth to show that no field experiment with the reported design and plot dimensions was visible at the claimed time and location (in Yazd). Despite providing valid examples of other fields, including my own, the authors rejected the presented evidence. However, they altered the experiment location in the 2020 paper to Shiraz University without referencing the 2017 paper.

Forage yield and quality as affected by salt stress in different ratios of *Sorghum bicolor*-*Bassia indica* intercropping

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ABSTRACT

Salinity tolerance is presumed to be improved in the intercropping system, however, there has been inadequate evidence to scientifically prove this. In the current study, the effect of salinity was examined on forage yield and quality of sorghum and kochia in an intercropping system with different planting ratios **in a two-year field trial at Shiraz University**. The treatments included three salinity levels of irrigation water; 2 (non-saline as a control), 7 and 14 dS m⁻¹ in main plots and planting ratios; sole sorghum (SS), 2/3 sorghum-1/3 kochia (S₂K₁), 1/2 sorghum-1/2 kochia (SK), 1/3 sorghum-

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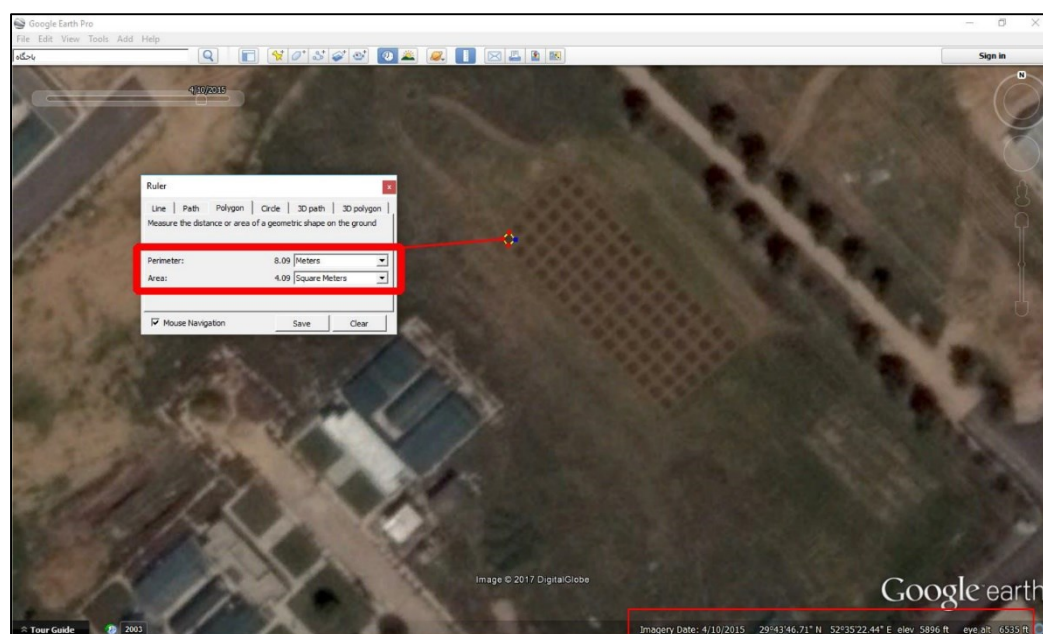
KEYWORDS

acid detergent fiber; crude protein; digestible nutrients; halophyte; mixed cropping; neutral detergent fiber

Materials and methods

In a **two-year field trial**, the influence of different intercropping systems was evaluated on forage yield and quality of sorghum and kochia under varying salinity levels **at Shiraz University, during 2015 and 2016 growing seasons**. Physico-chemical attributes of the experimental field soil are depicted in Table 1. The experimental set-up was a split-plot randomized complete block design with three replications. The treatments comprised irrigation water with varying levels as 2, 7, and 14 dS m⁻¹ in the main plots and five levels of planting ratios: sole sorghum (SS), 2/3 sorghum-1/3

Figure 1. Authors have explicitly reported the location of the field experiment as Shiraz University, in the second paper (2020).



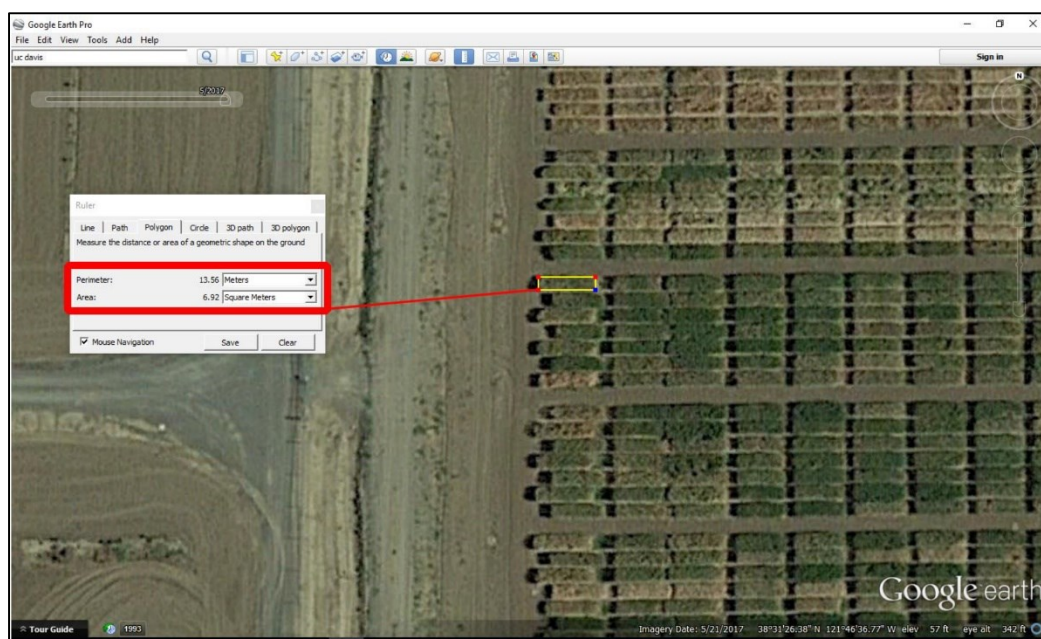


Figure 2: An example of capability and validity of Google Earth images in monitoring the experimental plots.

This falsification is a clear breach of scientific integrity, especially since the time and location of an experiment are the most critical factors in agricultural research. The Committee of Ethics at Shiraz University implicitly acknowledged that the experiment was not conducted at Shiraz University but acquitted the authors. Despite assurances from the publisher (Taylor & Francis) to follow up, no further response has been received, and the paper remains neither retracted nor revised. I have also shared my concerns on PubPeer (<https://pubpeer.com/publications/96479A22EEDD4C23B5D6DA1AE10DA3#1>) without any response from the authors.

References for the first case:

Hedayati Firoozabadi, A., Kazemeini, S. A., & Pirasteh Anoshe, H. (2017). Evaluation of different planting ratio of sorghum-kochia intercropping in varied salinity conditions. *Scientific Journal Management System*, 24(3), 685-698. doi:10.22092/ijrdr.2017.114014

Hedayati-Firoozabadi, A., Kazemeini, S. A., Pirasteh-Anosheh, H., Ghadiri, H., & Pessarakli, M. (2020). Forage yield and quality as affected by salt stress in different ratios of Sorghum bicolor-Bassia indica intercropping. *Journal of Plant Nutrition*, 43(17), 2579-2589. doi:10.1080/01904167.2020.1783301

2nd Case

Phantom Equipment: The Fictitious Lab Tools Scandal

In the second case, the same team (i.e. with the same supervisors) has published several papers involving various types of scientific misconduct. Notably, several laboratory or field equipment items have been reported in the materials and methods sections of these papers, which are not available at Shiraz University. I have reported the list of claimed tools on PubPeer, to the journal, and to the Committee of Ethics in Science of Shiraz University, requesting the serial numbers, registration codes, and/or images of these tools. However, no response has been received. The reported tools include:

1. Portable photosynthesis system (Li6400; LI-COR, Lincoln, NE, USA) coupled with a standard red/blue LED broadleaf cuvette (6400-02B, LI-COR) and a CO2 mixer (6400-01, LI-COR)

2. Portable infrared gas analysis system (IGRA model LCA4-ADC, Hoddeson, UK)
3. Mario MAX CN elemental analyzer (Elementar Analyser Gmb)
4. Quantum line sensor (Skye Instruments)

leaf above ear leaf), avoiding the midrib. Net photosynthetic rate was measured with a portable photosynthesis system (Li6400; LI-COR, Lincoln, NE, USA) coupled with a standard red/blue LED broadleaf cuvette (6400-02B, LI-COR) and a CO₂ mixer (6400-01, LI-COR), under a consistent controlled-light intensity of 1600 $\mu\text{mol m}^{-2} \text{s}^{-1}$. The measurements were ob-

determine total N concentration using a Vario MAX CN element analyzer (Elementar Analysensysteme GmbH). Based on the area, dry weight, and N content of the leaves used for PN measurements, specific leaf area ($\text{cm}^2 \text{g}^{-1}$) was calculated as leaf area (cm^2) per dry weight (g^{-1}), specific leaf N (SLN; g N m^{-2}) was calculated as N content (g) divided by whole-leaf area (m^{-2}). Photosynthetic N-use efficiency ($\mu\text{mol CO}_2 \text{g}^{-1} \text{N s}^{-1}$) was calculated as P_N/SLN . The P_N by SLN curve for leaves was fitted by applying a logistic equation (Sinclair and Horie, 1989). Instantaneous photosynthetically active radiation (iPAR; $\mu\text{mol m}^{-2} \text{s}^{-1}$) distribution in the canopy was measured at 40 day after sown (DAS) in 2014 and 2015, using a PAR Line Quantum Sensor (Skye Instruments Ltd). The measurements were conducted between 09:00 and 12:00 in three representative rows in each of the experimental plots. Each measurement was repeated three times. The probe was positioned horizontally between two rows,

surface and separated into leaves, stalk (leaf sheaths, tassels, husks, and cobs), and total ear fresh mass (FM). Results are presented on a fresh mass basis as this is most relevant for this crop. Harvest index was calculated as the ratio of ear yield over fresh weight per plant to total plant fresh biomass. All samples were dried at 105°C for 30 min, dried at 70°C to a constant weight, weighed to obtain the dry weight, and then ground into a fine powder. A portion (80 mg) of leaf tissue was used to determine total N concentration using a Vario MAX CN elemental analyzer (Elementaranalysen-systeme GmbH). The N-utilization efficiency (NUE) was calculated as ear FM per unit of N uptake at maturity (Moll et al, 1982). The N nutrition index was calculated as follows: N nutrition index = N concentration of the plant/ the critical N concentration for sweet corn. The critical N concentration (g kg^{-1}) was calculated by $34.0 \times \text{the dry matter (Mg ha}^{-1}) - 0.37$ as suggested by Plénet

Figure 3. Authors have claimed utilizing the laboratory equipment which were not available.

These tools were claimed to be used in the following papers:

Jafarikouhini, N., Kazemeini, S. A., & Sinclair, T. R. (2020). Sweet corn nitrogen accumulation, leaf photosynthesis rate, and radiation use efficiency under variable nitrogen fertility and irrigation. *Field Crops Research*, 257, 107913. doi: <https://doi.org/10.1016/j.fcr.2020.107913>

Jafarikouhini, N., Kazemeini, S.A., Ghadiri, H., & Lagrimini, M. (2016). Examination of photosynthetic nitrogen use efficiency of field-grown sweet corn (*Zea mays* L var merit) under water and nitrogen stress. *Maydica*, 61, 1-11. (Available as open access at: <https://journals-crea.4science.it/index.php/maydica/article/view/1554>)

Additionally, in the paper by Jafarikouhini et al. (2020), it was claimed that soil water content was measured using a neutron probe. Based on my investigations, such a device was not available to the authors. At Shiraz University, only the Department of Water Engineering has neutron probe, and its use requires the installation of aluminum tubes (for each experimental plot) and operation by certified technicians registered with the Atomic Energy Organization of Iran. No such tubes were planted, nor was the device used in the claimed field. Contradictorily, the dissertation's materials and methods section states that soil water content was measured using the gravimetric method. Indeed, three different methods for measuring soil water content have been reported in the mentioned paper and the corresponding dissertation: Time-domain reflectometry (TDR), neutron probe, and gravimetric method. Notably, in a single paper (Jafarikouhini et al., 2020), the authors claimed to have used two of these methods (i.e., TDR and neutron probe) simultaneously, raising questions about the consistency of the reported methodology.

25 % just before the silking stage. The plots were also fertilized at sowing with 150 kg P ha⁻¹ as triple superphosphate and 100 kg K ha⁻¹ as 60 % potassium sulphate.

Three water regimes were imposed using a drip irrigation system after seedling establishment based on estimates of extractable soil water. Soil water content was measured between rows in each plot using time-domain reflectometry rods (top 0.2 m of soil) and a neutron-probe reading down to 1 m depth. Irrigation was applied when the water in the soil profile of the wettest plot decreased to about 30 % of total extractable soil water. The fully watered plot received sufficient water calculated to return the soil to 100 % of field capacity (I100). In addition, two irrigation-deficit treatments were established by applying 80 % (I80) or 60 % (I60) of the water applied to the I100 treatment.

2.2. Leaf gas exchange

3.3. Soil water monitoring and irrigation

Before each irrigation event, the soil samples from I₁₀₀ (full irrigation) treatments were taken at 30 cm intervals down to 60 cm using post-hole auger from 4 spots at the center of each plot in three replications. The volumetric water content of the soil layer was measured using the gravimetric method and the depth of irrigation for I₁₀₀ treatments was calculated using the following equation:

$$D = \sum_{i=1}^n (\theta_{fci} - \theta_i) \Delta Z_i \quad (2)$$

Figure 4. Authors have claimed using controversial methods for soil water content measurements. The first and second images are taken from the paper and the corresponding dissertation, respectively.

Furthermore, the 2020 paper (published in Field Crops Research) claimed that data on global incident radiation was collected from a nearby weather station. The only weather station in the region, located at the School of Agriculture, Shiraz University, had its respective sensor out of service during the reported seasons. Despite requesting the claimed recorded data from the authors and Shiraz University, no response was received.

The interception of radiation by the leaf canopy was measured at midday (11:00 to 14:00) using a quantum line sensor (Skye Instruments) placed at the soil surface diagonally between adjacent rows. The measurements of fraction intercepted radiation were used to calculate by interpolation the fraction interception on each day of each plot. The global incident radiation was measured at a nearby weather station. Daily intercepted radiation was summed to obtain cumulative intercepted radiation on each day through the growing season. RUE was calculated for each treatment by linear regression using the seven days of harvested plant mass vs. cumulative intercepted radiation on each of the days.

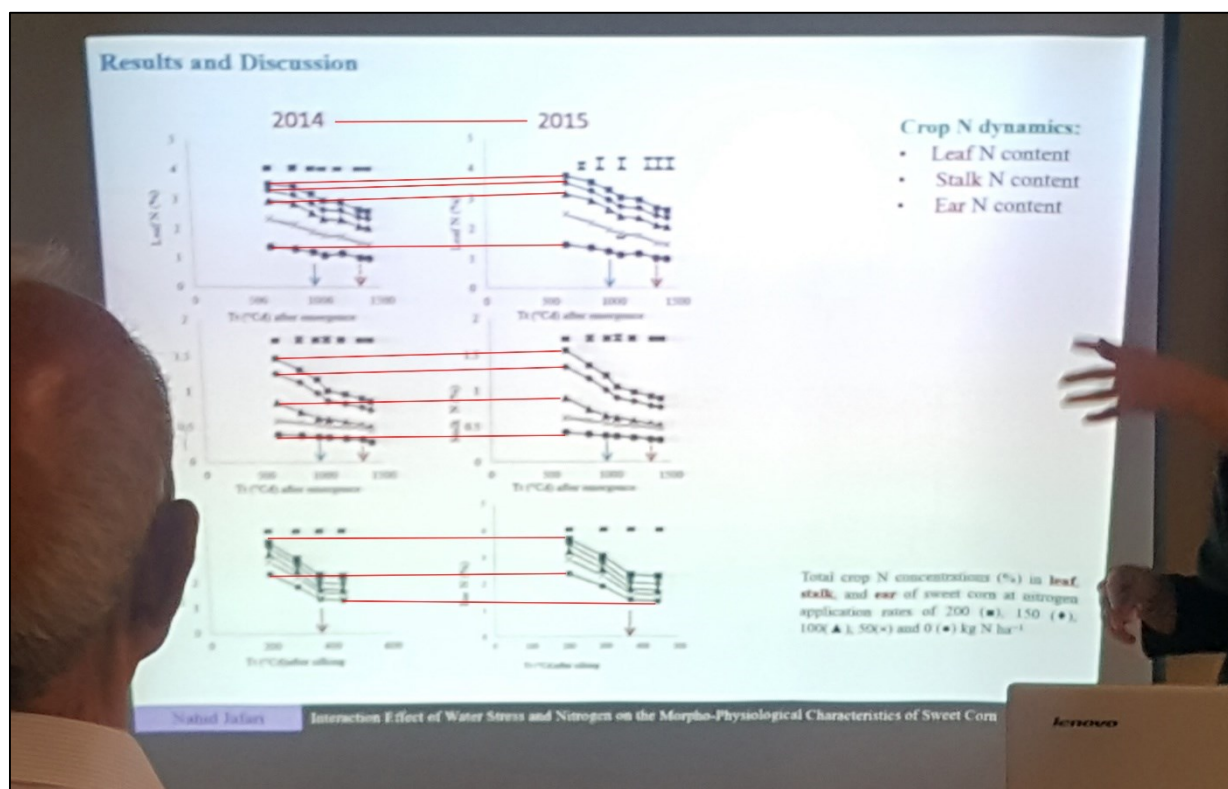
3. Results



Figure 5. Despite the claim of paper, the sensor of device for measuring the global incident radiation in the reported weather station was out of service during the claimed seasons.

Endless Duplications

Beyond the fraudulent reporting of equipment usage, significant evidence of data duplication was observed in the published papers and dissertation. These duplications were first noted during the defense meeting, where I questioned the student and the defense committee about the duplicated figures. Despite my inquiries, no clear response was provided. The duplicated figures and fabricated data were subsequently published in the PhD dissertation and at least two papers, including the 2020 paper in *Field Crops Research* (Elsevier).



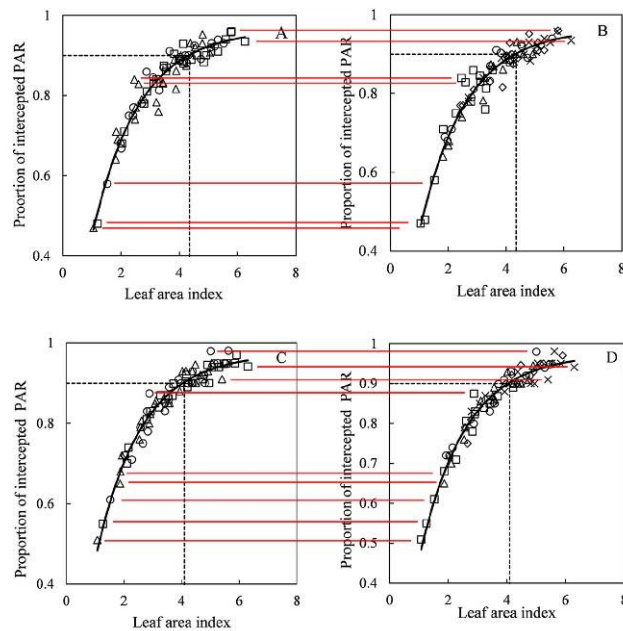
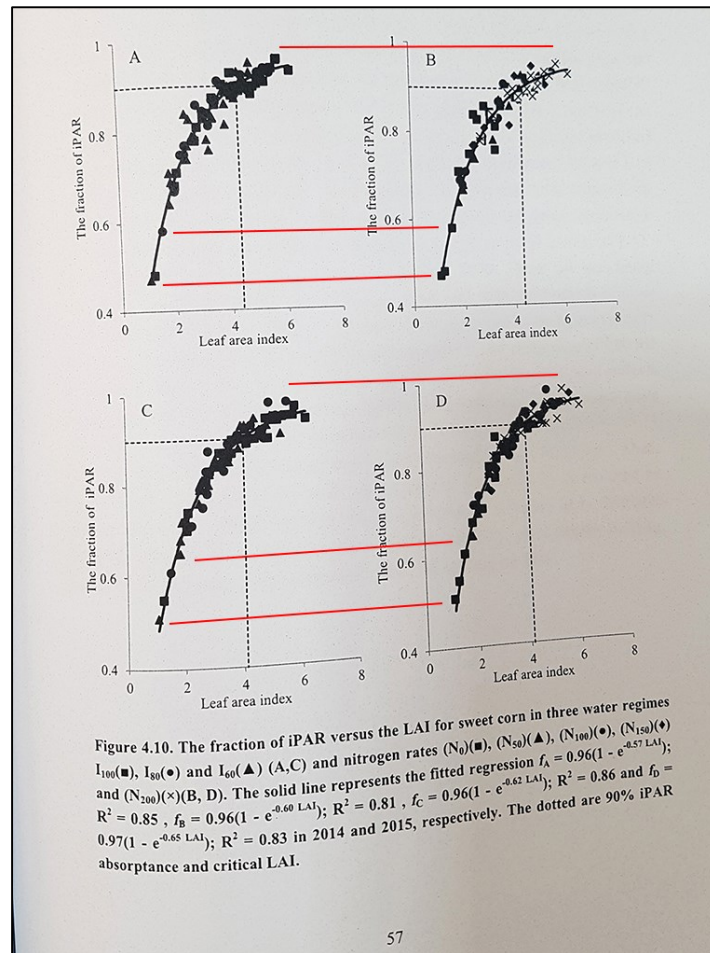


Figure 3 - Proportion of intercepted photosynthetically active radiation versus the green area index for sweet corn in three water regimes I_1 (■), I_2 (●), and I_3 (▲) and nitrogen fertilizer applications N_1 (■), N_2 (▲), N_3 (●), N_4 (◆) and N_5 (×) (A, B)(C, D) in 2014 and 2015. The solid line represents the fitted regression $y = 0.96(1 - e^{-0.62x})$; $R^2 = 0.96$, $y = 0.97(1 - e^{-0.64x})$; $R^2 = 0.95$ and the dotted 90% PAR absorbance and critical green area index, respectively.

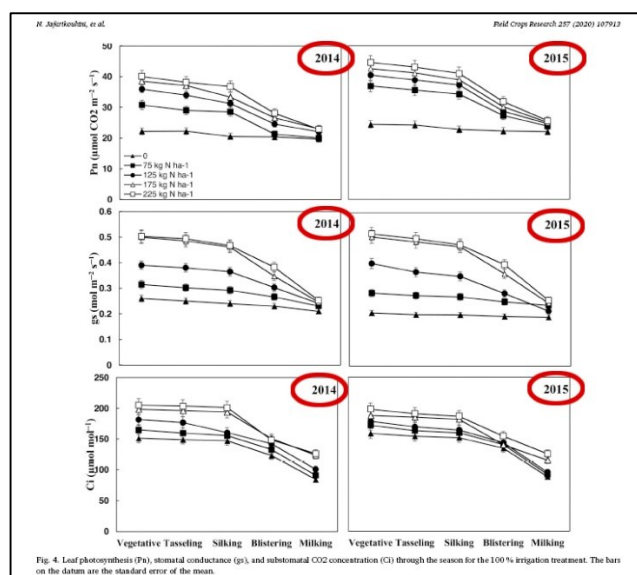


Fig. 4. Leaf photosynthesis (P_n), stomatal conductance (g_s), and sub-stomatal CO_2 concentration (C_i) through the season for the 100% irrigation treatment. The bars on the datum are the standard error of the mean.

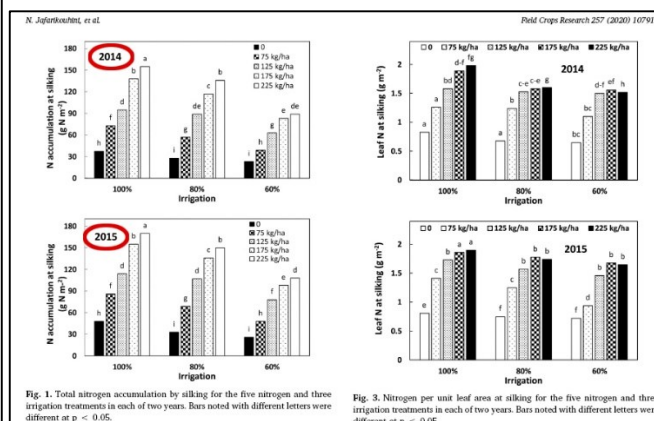


Fig. 3. Nitrogen per unit leaf area at silking for the five nitrogen and three irrigation treatments in each of two years. Bars noted with different letters were different at $p < 0.05$.

Figure 6. Examples of duplicated data in the figures of the study, including in the presentation of defense meeting in 2017, in the dissertation manuscript, and also in the paper (Jafarikouhini et al., 2016, and 2020). Red lines are drawn by me. For more examples, please see the respective link on PubPeer.

I reported these instances of misconduct to the journal and the Committee of Ethics in Science at Shiraz University. Although the Field Crops Research journal conducted an investigation, they concluded that the findings were novel, and the similarities in figures were not considered duplications. As a result, the paper was not retracted. I also reached out to the corresponding author of another paper (Jafarikouhini et al., 2016) from the University of Nebraska-Lincoln, but received no response. Similarly, Shiraz University acquitted the authors without detailed explanation. The authors went on to publish two more papers from this compromised study in journals of Taylor & Francis, but my warnings to the publisher were unsuccessful.

The full list of papers derived from this study includes:

Jafarikouhini, N., Kazemeini, S.A., Ghadiri, H., Lagrimini, M., (2016) Examination of photosynthetic nitrogen use efficiency of field-grown sweet corn (*Zea mays* L var merit) under water and nitrogen stress. *Maydica*, 61, 1-11.

Jafarikouhini, N., Kazemeini, S. A., & Sinclair, T. R. (2020). Sweet corn nitrogen accumulation, leaf photosynthesis rate, and radiation use efficiency under variable nitrogen fertility and irrigation. *Field Crops Research*, 257, 107913. doi: <https://doi.org/10.1016/j.fcr.2020.107913>

Jafarikouhini, N., Kazemeini, S.A., Sinclair, T. R. (2020). Sweet Corn Ontogeny in Response to Irrigation and Nitrogen Fertilization. *Journal of Horticulture and Plant Research*, 10, 23-29.

Jafarikouhini, N., Kazemeini, S. A., & Sinclair, T. R. (2021). Fresh sweet corn yield sensitivity to deficit nitrogen and water conditions. *Journal of Crop Improvement*, 1-11. doi: 10.1080/15427528.2021.1995560

The Role of External Corresponding Authors

It should also be emphasized that in these two misconduct cases, besides Shiraz University, three other universities have been inevitably included: University of Nebraska – Lincoln (Mark Lagrimini), North Carolina State University (Thomas R. Sinclair), and The University of Arizona (Mohammad Pessarakli). The manner in which these papers were published involved researchers at Shiraz University sending their claimed results to the

corresponding authors at these other universities, who then submitted the manuscripts to journals. This was done without any responsibility to ensure the experiments or measurements were actually conducted or even to check the manuscripts. This approach raises concerns about the responsibility and accountability of corresponding authors in ensuring the integrity of submitted research."

Questionable Defense by Colleagues

In response to the concerns I published on PubPeer regarding the second study, Thomas Sinclair, the corresponding author of the 2020 paper in *Field Crops Research*, claimed that he had received a letter from the authors and others at Shiraz University rejecting my concerns. The letter stated:

"...The research was undertaken as described in our paper. We completely reject the unsupported claims that the paper is fraudulent and unethical. [Signature:] Dr. Nahid Jafarikouhini, North Carolina State University; Dr. Abdolreza Kazemeini, Shiraz University; Dr. Hossein Ghadiri, Shiraz University; Dr. Mohsen Edalat, Shiraz University; Dr. Ali Dadkhodaie, Shiraz University; Dr. Thomas R. Sinclair, North Carolina State University."

I requested Dr. Sinclair to publish the original version of the letter with the authors' signatures; however, he refused. This questionable defense from colleagues further complicated efforts to uncover the truth. I wonder what their opinion is about the undeniable and various pieces of evidence of scientific fraud.

Lesson One

The Committee of Ethics in Science at Shiraz University refused to announce its official opinion on the two scientific misconduct files until I involved the Commission of Education and Research of the Parliament of Iran. Eventually, after more than a year, the committee announced the acquittal of the authors. Facing this unacceptable judgment, I published the story of these scientific violations and the detailed evidences on X (formerly Twitter) under the title "Lesson One." In Lesson One (in Farsi), I concluded that Shiraz University appears to have been profoundly impacted by systematic issues that hinder its ability to address scientific misconduct effectively. Notably, even evidenced warnings before the publication of these scientific papers were unhelpful.

Challenges Faced by Shiraz University in Addressing Misconduct

Following the dismissal of evidence of scientific violations by Shiraz University's Committee of Ethics in Science and various publishers, I sought help from other responsible organizations, including the Ministry of Science, Research and Technology of Iran, the General Inspection Office, and the Supreme Council of the Cultural Revolution. Despite sending many letters and making numerous contacts over the past three years, either no response was received, or they merely conducted a superficial inquiry, receiving a brief response from Shiraz University stating, "we did not find any violation."

In another attempt, I used the IranFOIA system (established under the new law for publishing and free access to information in Iran) to legally compel Shiraz University to disclose the requested information, such as the exact coordinates of the location of the experimental field in the first case and the details of the claimed laboratory equipment and data of global incident radiation in the second case. Unfortunately, Shiraz University did not publish the requested information. Subsequently, I filed a criminal complaint against Shiraz University related to this law. After about nine months, the court acquitted Shiraz University and accepted its defense for avoiding

the publication of the requested information. Official objections to the court's ruling and requests for retrial were also unsuccessful.

Sidelined, but not Silenced: Punishment of the Whistleblower

(I am writing these lines to share the experiences gained during my pursuit of these misconduct cases, not merely to recount a personal story.)

After publishing Lesson One and several other posts on X, I encountered the following obstacles:

Direct Issues:

1. Shiraz University filed a criminal complaint against me for my claims about scientific misconduct. However, unable to prove the crime, the university eventually dropped the complaint.
2. Alireza Hedayati Firouzabadi, the student and author in the first case, filed another criminal complaint against me, but I was acquitted in this case as well.

Indirect Issues (currently, I cannot directly attribute these to my pursuits; however, they occurred):

3. Shiraz University refused to issue the original version of my two-year postdoctoral certification, even when I offered to waive the postdoctoral salary.
4. Shiraz University rejected my application for admission to the national call for faculty recruitment in 2023. Despite my attempts to challenge this decision legally, my protest was frozen in the Ministry of Science.
5. Despite the importance and necessity of the topic, Shiraz University rejected my research proposal on Crop Phenotyping. According to my information, at least one referee and decision-making official was a member of the Committee of Ethics in Science at Shiraz University, and another had previously given false testimony in scientific fraud cases (see Dr. Sinclair's letter).

Last word

The authors, professors, and officials involved in these cases at Shiraz University disregarded my warnings, perhaps assuming that I would eventually graduate and let the matter go. However, my commitment to academic integrity and to Shiraz University remains steadfast. Shiraz University is not just an institution—it is our academic home, and restoring its integrity is a shared responsibility. Through persistence and collective effort, I believe we can overcome the challenges posed by systematic misconduct and revive the values that underpin true scholarship.

(All unpublished evidence and documents are available upon request. Additionally, future updates on this story may be shared via @haqueshenas on X)

November 2024